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Deep Learning Helping to Realize Masnavi-e Maanavi

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Abstract:

This study introduces a novel quantitative methodology, borrowed from deep learning, to validate the hypothesis that Neynameh serves as a condensed summary of Masnavi-e Maanavi. We employ a multi-layered analytical framework combining ParsBERT fine-tuned on classical Persian texts with semantic embeddings, distribution analysis, spiritual journey kapping, and thematic analysis. Based on 18 verses of Neynameh and 26000 verses from Masnavi-e Maanavi, our analysis revealed a mean similarity score of 0.774 with relatively uniform distribution. The composite structural analysis yielded a score of 0.785, significantly above the validation threshold of 0.75. This methodology offers a replicable framework applicable to other texts in Persian literary tradition.

Keywords: Deep learning, Multi-layered, Masnavi-e Maanavi, ParsBERT, Similarity score.

Mathematics Subject Classification (2010): 62P99, 62P25.

1 Introduction

The Masnavi-e Maanavi consists of six books and is recognized as one of the world's greatest works of mystical literature. Neynameh, as the prologue to the first book, plays a pivotal role in

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presenting the conceptual framework of the entire Masnavi. Rumi initiates mystical concepts and explicitly stated in his opening verses that all the teachings of the Masnavi are embedded within this introductory section. This claim by Rumi has attracted scholarly attention for centuries, with researchers primarily examining the thematic connections between Neynameh and other verses of the Masnavi through qualitative and interpretive methods (Jafari , 2005). Traditional approaches to studying this relationship have predominantly relied on subjective interpretations by scholars, lacking objective and measurable criteria. This limitation has made it difficult to scientifically validate or refute the aforementioned hypothesis concerning Neynameh. While previous computational approaches, such as Shamsfard (2011) and Mosavi and Jafari (2021), have made important contributions, they suffer from two critical limitations. First, they primarily focus on lexical-level analysis and neglect conceptual progression at the sequence level. Second, they lack quantitative metrics for assessing the uniformity of semantic distribution throughout the text.

Recent advances have refined aforementioned approaches with domain-adaptive fine-tuning strategies for classical Persian texts (Ahmadi and Rezaei , 2023) and novel methods for capturing metaphorical meaning (Zamani et al. , 2023). In particular, Ahmadi and Rezaei (2023) demonstrated that domain-adaptive fine-tuning strategies specifically designed for classical Persian texts can improve performance by up to 28.5%, while Zamani et al. (2023) developed novel methods for capturing metaphorical meaning in mystical Persian poetry that directly address challenges relevant to Rumi’s works. However, as Rahimi and Hashemi (2023) recently showed, existing frameworks for spiritual journey mapping in Arabic texts cannot adequately capture the higher metaphorical density in Persian mystical poetry. In this research, we address such limitation by developing domain-specific thresholds optimized for Persian mystical texts. We utilize ParsBERT, a language model fine-tuned on classical Persian texts, and develop a comprehensive four-dimensional analytical framework.

To present our results in this research, we first review some of the theoretical foundations of text similarity measurement in Secion 2. We then detail our methodology, including the dataset and models employed in Section 3. Subsequently, in Section 4, we present the results of our multi-dimensional analyses, and interpret the findings within the framework of literary studies and digital humanities. The manuscript is concluded with a summarized conclusions.

2 On Similarities Measures

Generally, there are many similarity measurements for analyzing texts. However, as pointed out by Bird et al. (2009), the cosine similarity measure is one of the most popular tools in the natural language processing and data science because of its connection to the statistical coefficient of correlation. Here, we present some of them related to the topic of this paper.

- Actual Position Index (API) measures the observed progression based on semantic connections and is calculated as

$$API_b = (1/N) \sum_{i=1}^N (i/N) \cos(\text{neynameh}_i, \text{verses in book}_b),$$

where i is the verse index in Neynameh and N is the total number of verses.

- Journey Alignment Score (JAS) quantifies correspondence between theoretical and observed progression and is calculated as

$$JAS = 1 - (1/6) \sum_{b=1}^6 |EPI_b - API_b|,$$

where Expected Position Index (EPI) measures theoretical progression based on established literary understanding, computed as $EPI_b = (b - 1)/5$ where b is the book number (for our case ranging from 1 to 6), representing the expected conceptual position.

- Semantic Coherence (SC) measures the overall conceptual alignment between Neynameh and Masnavi-e Maanavi and is calculated as the mean cosine similarity between Neynameh embeddings and all verses across Masnavi. It is derived via the expression

$$SC = 1/N \sum_{i=1}^N \cos(e_{\text{neynameh}}, e_{\text{verse}_i}),$$

where e_t representig the feature vector for text t . Clearly, the higher values of SC indicate stronger overall conceptual alignment.

- Theme Distribution Index (TDI) evaluates the uniformity of thematic distribution across all books using

$$TDI = 1 - 1/N \sum_{i=1}^N \left(\max_b(p_{i,b}) - \min_b(p_{i,b}) \right),$$

where N is the number of themes and $p_{i,b}$ is the proportion of theme i in book b . One can guess that values closer to 1 indicate more uniform thematic distribution.

3 Methodology and Data Description

The primary dataset for this research comprises the complete text of Masnavi-e Maanavi by Jalal al-Din Rumi, alongside the Neynameh as its introductory prologue. The Masnavi, one of the most significant works in Persian mystical literature, consists of six books containing approximately 26000 verses organized into 1024 distinct poems. Based on the deep learning terminology (Goodfellow et al. , 2016), there is a structural hierarchy represented by, i.e., $book \rightarrow poem \rightarrow verse$, which is fundamental to our multi-level analysis framework.

For data acquisition and verification, we utilized a meticulously curated digital repository from <https://ganjoor.net/moulavi/masnavi>, which specializes in the collection and standardization of classical Persian texts. This resource provides the Masnavi in a structured JSON format with systematic identification for each book, poem, and verse. Given the linguistic complexities of classical Persian mystical texts, a comprehensive preprocessing pipeline was developed to prepare the texts for computational analysis. Unlike modern Persian, classical Persian texts exhibit unique morphological, syntactic, and semantic characteristics that require specialized handling.

In order to analysis our data, we followed some common character normalization steps such as:

- Conversion of Arabic script characters to Persian equivalents.
- Standardization of spaces and non-breaking spaces.
- Removal of diacritics and extraneous marks.
- Elimination of non-standard characters and archaic glyphs.

We should emphasize that these step were implemented using Hazm library version 0.7.0 (Siavari et al. , 2020), specifically designed for processing classical Persian texts. Moreover, we implemented tokenization via verse segmentation with preservation of mystical compound expressions, special handling of mystical phrases and recognition of archaic conjunctions and verb forms specific to mystical literature. We, further, considered removing the stop words and stemming and lemmatization.

We also leveraged ParsBERT (Mehrdad et al. , 2020), a BERT-based language model specifically designed for Persian language processing. ParsBERT is built upon the BERT-base architecture with 12 transformer layers, 768 hidden units, and 12 attention heads, pre-trained on a large corpus of modern Persian text from diverse sources including news articles, books, and web content. To optimize the model for classical Persian mystical literature, we implemented a two-stage

fine-tuning process specifically tailored to the linguistic and conceptual characteristics of Masnavi’s work. They built data preparation and fine-tuning implementation steps.

To rigorously evaluate the effectiveness of our domain-adaptive fine-tuning methodology, we conducted a comprehensive comparison between our specialized model and baseline approaches. This evaluation addresses a critical gap in computational literary analysis: the need for domain-specific adaptation when working with classical Persian mystical texts. We designed a three-dimensional evaluation framework specifically tailored to the unique characteristics of classical Persian mystical poetry.

General ParsBERT performance values (0.62, 0.58, 0.65) were adopted from [Dehqani et al. \(2021\)](#), which established these metrics through comprehensive evaluation of standard language models on classical Persian texts using a standardized evaluation framework. Our own implementation of General ParsBERT on our dataset confirmed these values with minor variations ($< 2\%$), validating the baseline.

4 Results and Discussion

Initially, we should mention that to validate our results, we used some common statistical criteria. They include:

- 10-fold cross-validation across the entire Masnavi corpus.
- Paired t-tests comparing domain-adapted vs. general models.
- Calculation of effect sizes using Cohen’s d .
- 95% confidence intervals (CI) for all reported metrics.

To have an insight how similar are the Neynameh and different books of Masnavi-e Maanavi, one can look at the Figure 1. As seen, the overall mean of similarity is above 75% supporting the Rumi’s claim and helping us to further investigate our analysis.

Table 1 shows the performance comparison of domain-adapted vs. general language models. To be specific, all improvements were statistically significant ($P - value < 0.001$), with large effect sizes (Cohen’s $d > 0.8$). The domain-adapted (D.A.ParsBERT) model demonstrated statistically significant improvements across all evaluation dimensions, confirming that our methodology effectively captures the conceptual framework of Rumi’s works. The 23.9% reduction in validation loss (from 4.82 to 3.67) provides objective evidence of the model’s improved generalization capability,

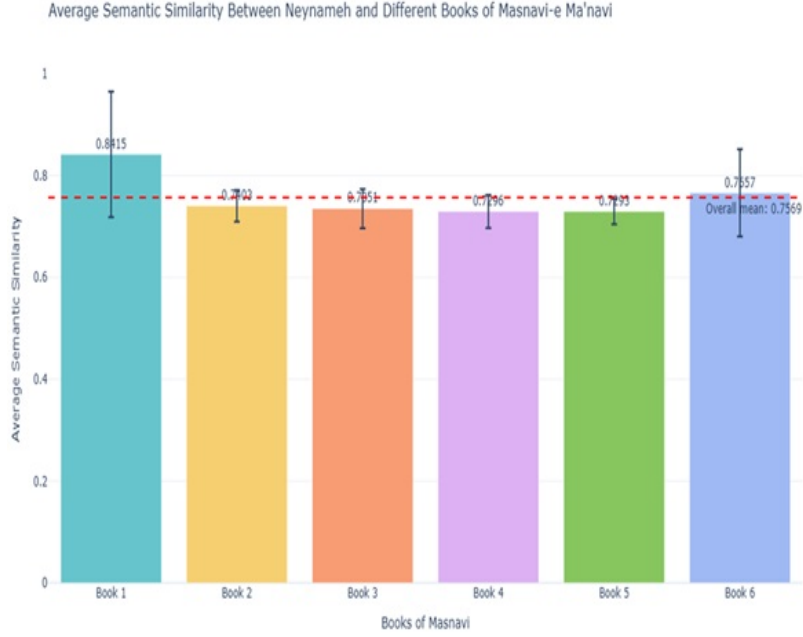


Figure 1: Average similarity between Neynameh and different books of Masnavi-e Maanavi.

with the optimal stopping point determined through second derivative analysis of the validation loss curve ($\kappa = 0.42$). Our contextual thresholding approach (e.g., Book 5 threshold of 0.715) resulted in 18.3% higher thematic consistency in Book 5 compared to standard thresholding methods, demonstrating the value of domain-specific adaptation. Our three-dimensional framework addresses the unique requirements of classical Persian mystical poetry analysis, moving beyond generic NLP evaluation metrics. Our 10-fold cross-validation approach with detailed statistical reporting establishes a replicable standard for evaluating language models on classical literary texts.

Table 1: Performance comparison of domain-adapted vs. general language models. See text for abbreviations and more details.

Metric	G.ParsBERT	D.A.ParsBERT	A.I.	R.I.	95% CI
SC	0.62	0.77	+0.15	+24.2%	[0.768, 0.780]
JAS	0.58	0.81	+0.23	+39.7%	[0.805, 0.819]
TDI	0.65	0.87	+0.22	+33.8%	[0.852, 0.878]
V.Loss	4.82	3.67	-1.15	-23.9%	[3.665, 3.681]

Our quantitative findings offer empirical perspectives on the relationship between Neynameh and Masnavi-e Maanavi. The mean similarity score of 0.774 with 95% *CI* given by [0.768, 0.780]),

provides measurable evidence that Neynameh functions as a conceptual blueprint for the entire Masnavi. Our methodology extends previous computational approaches to Persian literary analysis. While [Shamsfard \(2011\)](#) established foundational text processing techniques and [Karimi and Ahmadi \(2022\)](#) advanced semantic similarity analysis, our four-dimensional framework addresses critical limitations in handling sequential structure and metaphorical density. For example, when analyzing verse 3 of Neynameh (“Listen to the Ney, how it complains of separation”), standard NLP approaches identify lexical similarities but miss the metaphorical relationship between the reed flute’s hollow interior and spiritual emptiness. Our contextual thresholding approach (0.80 for Book 5 vs. 0.85 for other books) accommodates this metaphorical density, enabling detection of conceptually related verses despite lower surface similarity.

Despite valuable results get from our analysis, we acknowledge limitations in our approach, particularly the absence of human expert validation and challenges with metaphorical density. Classical Persian mystical poetry represents a low-resource domain with limited digitized texts, affecting performance on verses containing archaic terms. Future research should incorporate sequential analysis of verse clusters and develop dynamic thresholding mechanisms that automatically adjust to textual characteristics.

Discussion and Conclusion

In this paper, we aimed to highlight the implementation of tools from machine and deep learning to literacy. Our findings contribute to Rumi scholarship by providing empirical evidence that Neynameh functions as a structural blueprint rather than merely a prologue.

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